



Real-Time Cyber-physical Systems: State-of-the-Art and Future Trends

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Abstract

The real-time cyber-physical systems (CPSs) describe a wide range of complex and interdisciplinary engineered systems that integrate real-time embedded computing and control technologies (cyber component) into physical world. To further understand real-time CPSs, this chapter presents a brief survey of the rele-

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vant work, exploring the emergence of real-time CPSs, the system requirements, the relation to other research areas, similar and prevalent concepts, and important practical applications. In addition, this chapter lists several important technical challenges and elaborates insightful concepts by utilizing specific applications. Real-time CPSs have various applications spanning different scales due to their very broad research areas. The future and next-generation technologies are anticipated to have a significant impact on real-time CPS applications. Hence, all real-time CPS applications are required to be developed and designed taking into account advanced technologies, essential system requirements, and overall influence on physical world.

Keywords

Real-timeliness · Real-time cyber-physical system (CPS) · Embedded computing technologies · System-level requirements

1 Introduction

Real-time and embedded systems are utilized in diverse application domains that need timely processing of huge amount of real-time data. Examples of real-time data include sensor data in sensor networks, positions of aircrafts in an air traffic control system, and temperature as well as air pressure in an engine control environment. Real-time data are usually stored in real-time database system (RTDBS). Such data are utilized to model the current status of components in the system. The major difference between the real-time data and traditional data stored in databases is that the real-time data have sampled values which are valid for only a certain time interval (Ramamritham 1993; Locke 1997; Stankovic et al. 1999).

The emergence of real-time systems is related to the evolution of the computer. The state-of-the-art real-time systems utilized in monitoring and controlling military aircraft weapons systems, or medical monitoring equipment, are complex and still demonstrate the features of the developed systems between the 1940s and 1960s (Laplante 2004).

In the last two decades, the Internet and embedded systems have been revolutionized, and many devices have been connected to the Internet. This increases the interaction of people with physical world. The advancement and the emergence of modern solutions in information technology resulted in modern intelligent solutions on cyber-physical systems (CPSs). CPSs integrate computing, communication, and storage capability with monitoring. CPSs offer advantages such as efficiency and safety. In addition, they allow individual entities to form complex systems with new capacities.

A state-of-the-art complex system which has interfaces with physical world and computing components running software with time-critical applications is an example of a CPS. CPSs are often safety-critical. That is if they miss time deadlines or in case of component failures, the consequences are life-threatening. Such systems are referred to as real-time CPSs. Hence, the real-time CPSs are governed

by strict timing deadlines. Based on the degree of criticality of the application and the task deadline, we can further categorize the real-time CPSs as hard, firm, or soft real-time CPSs. As some examples for such real-time CPSs, we can refer to antilock braking system (ABS) in automotive, temperature/pressure control systems in manufacturing plants, laboratory robotics, and flight controls in avionics.

This chapter first introduces the real-time systems and the various kinds of real-time systems. It then explains the real-time CPSs, their requirements, and important components. Some similar concepts to real-time CPSs are introduced and important applications are discussed. Finally, some challenges of real-time CPSs conclude this chapter.

2 Real-Time Systems

Real-time systems (RTS) are the computing (or information processing) systems which must respond to inputs within a specified time frame. As much as the correct processing and response is important, the time at which the results are generated is also essential. One simple example of a real-time system is a video game. In this RTS, the system receives the inputs from the user, and after processing, it updates the game state for the user and displays it on the screen. Another example is in the airport when a passenger has a flight departure in an hour from Melbourne to Sydney in Australia. The passenger approaches the airline reservation counter to collect his/her boarding pass. At the counter, the ticket agent inserts the passenger's information in the computer and the boarding pass is issued within seconds. Finally, consider the digital fly-by-wire flight control system with a computer as its main processing system. There are many control systems in the aircraft including control surfaces (e.g., rudders, ailerons, elevators, etc.). These control systems consist of variety of controllers and sensors. The computer receives the electronic signals from these control systems. These signals are processed by the computer, and the controlling signals are transmitted to the actuators (control surfaces) and the results further displayed on pilot screen. Too many delays in different control signal processing by the computer in different tasks might result in airplane instability and possible crash. The main common characteristic of the aforementioned examples is that the system must generate the output within a specified time frame. Clearly, failure to meet such a requirement results in a consequence. How severe the consequence is depends on the system. If the game player is slow, the user will lose interest in the game and there is no catastrophic consequence. However, if the fly-by-wire flight control system fails to meet many deadlines in response to the inputs, it can lead to the airplane crash.

Based on the consequences of the response time, RTS are categorized as (Kopetz 2011):

- **Safety-critical systems:** include all types of real-time systems which can lead to catastrophic consequences if the systems fail to function properly.
- **Hard real-time systems:** guarantee to meet the required deadlines in completion of all real-time tasks. Failure to meet a single deadline can lead to catastrophic

consequences such as physical damage or loss of life. Hard real-time systems are safety-critical systems.

- **Firm real-time systems:** allow the occurrence of a few number of missing deadlines. More than a specific number of missed deadlines will result in total system failure or catastrophic consequences.
- **Soft real-time systems:** prioritize the real-time tasks over non-real-time tasks. Low performance of the system is allowed with several number of missing deadlines; however, there are no failure (or catastrophic) consequences due to missing deadlines.

Among the aforementioned examples, the airport example and digital fly-by-wire flight control system are hard real-time systems, and video game is a soft real-time system. A simple example for firm real-time system is weather forecast system.

In the information age, the Internet has transformed the way we search for new information, we study, we run our business, and even we entertain. That is, very quick interaction with other people and obtaining our required information (in a very short time) is feasible using the Internet. However, there are still some gaps between the cyber world (i.e., the digital environment of information, computations, and computers) and the physical world (real-world networks and systems). Hence, with the aid of technological advances, a new generation of systems (e.g., self-optimizing transportation systems, energy efficient and sustainable airplane and automobile, etc.), which are called cyber-physical systems (CPSs), emerged. The CPSs introduce the integration of information and networking into physical objects. By integrating information and communications into such systems, a new generation of intelligent and autonomous systems could be developed. These kinds of systems fall in RTS category and termed real-time cyber-physical systems (RTCPs).

3 Real-Time Cyber-Physical Systems

Cyber-physical systems (CPSs) consist of cyber systems and physical systems. Cyber systems include software, embedded computation, or network systems with computation, communication, and control procedures. Physical systems are either natural or man-made systems which can be analyzed by the laws of physics and operate in continuous time domain. The example of physical systems are power grids, transportation networks (e.g., airlines and highways, etc.), factories, etc. Hence, the CPSs introduce a tight integration of communication and control, information, and computation into physical systems as shown in Fig. 1.

Definition 1. Real-Time CPS: A real-time CPS integrates computing, communication, and storage capabilities with monitoring and/or control of physical objects and must do so reliably, safely, securely, and efficiently and in **real time**, whose main characteristics are as follows (see Fig. 2) (Sanislav and Miclea 2012):

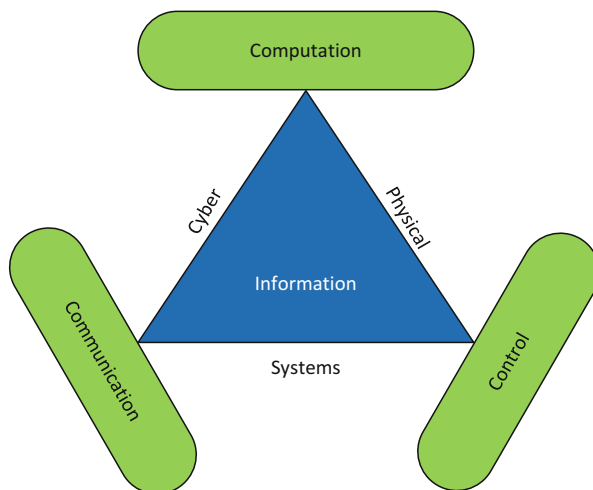
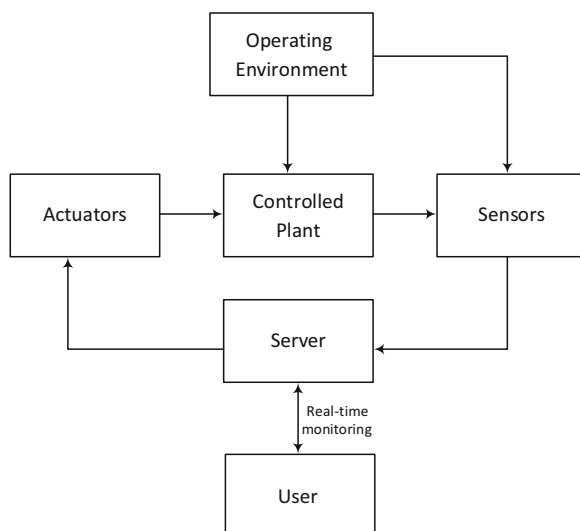


Fig. 1 Structure of cyber-physical system

Fig. 2 Structure of real-time cyber-physical system



- Communication of input and feedback from/to the physical environment, which requires secure channels
- Management and distributed control and monitoring
- Real-time performance requirements
- Wide geographical distribution without distributed physical security components
- Very large-scale control systems

The real-time CPSs are safe, efficient, and reliable, which makes them advantageous. They are hierarchical, distributed (networked) control systems with particular “self”-features, e.g., self-optimization to alleviate the effects of disturbance, self-configuration for manageability and resilience, etc. (Kovács házy 2018). Real-time CPSs have variety of applications including avionics, transportation, manufacturing processes, energy systems, healthcare, assisted living, social networking and gaming, agriculture, etc. The most recent applications of real-time CPSs are autonomous vehicles, smart infrastructure (i.e., smart cities), and Industry 4.0 (Kovács házy 2018; Lu et al. 2016).

Most of real-time CPS applications require the strict deadlines in performing real-time activities. For instance, the activities such as steering and acceleration/braking involved in an autonomous car must be performed with receiving real-time data with deadlines in the range of milliseconds for distributed architecture of modern cars.

As real-time CPS systems are time sensitive, to guarantee a deterministic behavior of CPSs, deterministic real-time networks with bounded delay and delay variance (jitter) are required. The examples of such networks are real-time Ethernet fieldbus solutions, e.g., EtherCAT, PROFINET IRT, SERCOS III, etc. However, none of them are standardized by IEEE and none of them are completely ready for real-time CPS applications (Kovács házy 2018). Hence, IEEE developed the IEEE 802 standard with real-time features under time-sensitive networking (TSN) standard, and this has been first started for IEEE 802.3, Ethernet networks. To achieve deterministic network delay and jitter for time-sensitive traffic, TSN aims to eliminate the nondeterministic queuing delays in network elements. TSN is analogous to a complex Ethernet fieldbus; however, it can provide the low-level, real-time, dynamic, virtualized, and distributed architecture for future wired CPS systems. Some features of TSN can also be utilized for IEEE 802.11 (Wi-Fi). The most important standards in TSN are:

- P802.1 AS, Revision, Timing, and Synchronization
- P802.1 Qcc, Stream Reservation Protocol
- IEEE 802.1Qbv-2015, Enhancements for Schedules Traffic
- IEEE 802.1CB-2017, Frame Replication and Elimination for Reliability
- IEEE 801.1 Qch-2017, Cyclic Queuing and Forwarding
- IEEE 801.1 Qci-2017, Per-Stream Filtering and Policing
- IEEE 801.1 Qbu-2016, Frame Preemption

One of the important applications of real-time CPSs is in Industry 4.0. For instance, wireless sensor-actuator networks (WSANs) for industrial control systems are one of the important types of real-time CPSs which have been emerged in the dawn of Industry 4.0 (Lu et al. 2016). Industry 4.0 or fourth industrial revolution introduced by German government in 2012 is mainly based on the application of real-time CPSs to manufacturing processes via high-tech strategies (Li and Lau 2018). This enables manufacturers to visualize the manufacturing processes including order fulfillment, mass production, machinery maintenance, machine

failure prediction, etc. Therefore, various electrical and electronic devices, e.g., personal computers, smartphones, smartwatches, machinery robotics, and enterprise resource planning (ERP) systems, are integrated together and communicated with each other to offer an optimal solution for improving productivity through the Internet. In fact, Industry 4.0 is the transformation of traditional plants into smart factories. This significantly impacts the value chain of companies and causes the disruption of traditional production in the factories. The combination of Internet of Things (IoT), digital services, and real-time CPSs utilizes the information and communication for the integration of physical, virtual, and digital worlds (Pisching et al. 2018). The manufacturing and production based on Industry 4.0 considers the products as smart components capable of communicating (the information by the integration of physical and digital systems) with shop floor machines to meet customer needs in on-demand production.

As the real-time CPSs are the important parts of the Industry 4.0 and industrial automation applications, it is necessary to investigate the requirements and components of CPSs, which will be studied in the next section.

4 Requirements and Components of Real-Time CPSs

Real-time CPSs integrate computing and communication capabilities with monitoring and control of entities in the physical world. Hence, these systems usually comprised a set of networked agents, e.g., sensors, actuators, control processing units, digital signal processing (DSP) units, and communication devices, which turn the real-time CPSs into complex systems and result in difficulties in the requirements engineering process. This is because the requirements are also conflicting, redundant, and complex, which increase the time and cost of developing the real-time CPS.

Elicitation phase is an essential part of the requirement engineering, due to its vital role in the whole development life cycle (Penzstadler and Eckhardt 2012). Elicitation phase is a collection of activities which include gathering information from stakeholders and classify those requirements based on priority into normal, expected, and exciting requirements. Normal requirements must be achieved by the developers. These types of requirement are explicit agreement between stakeholders and development team. Expected requirements are essential for the system to work properly. Exciting requirements are extra features to promote the complete satisfaction of users/clients.

The identification and refinement of the requirements for real-time CPSs are difficult tasks as real-time CPSs are complex and distributed systems. However, if the requirements are properly collected and refined, the development of such systems will be easier. Hence, the collection and refinement processes of the requirements for CPSs are essential parts of the system development. In general, based on existing literature, the requirements for developing real-time CPSs can be categorized into two major groups: time-related requirements and generic requirements.

- **Time-Related Requirements:** Time is a necessary requirement for development of real-time CPSs, which makes the system to be predictable, reliable, and deterministic. The author in Lee (2008) has studied the design challenges of real-time CPSs and has investigated the adequacy of the computing and networking technologies for developing real-time CPSs. In order to satisfy the time requirement of real-time CPS, Edward E. Lee has suggested to gradually improve the technologies and computer OS (operating system) architecture. CPSs need to deal with huge amount of data in a timely and secure fashion. The authors in Kang and Son (2008) have proposed an information-centric approach for timely and secure real-time data services in CPSs. In this approach, network-enabled real-time embedded databases (nRTEDBs) communicate (with each other and) with wireless sensors in a secure and timely manner. Hence, the overall timeliness, security, and efficiency of the system will be enhanced. The problem of optimizing the spatiotemporal distribution of real-time CPSs has been investigated in Kong et al. (2010). The authors have proposed an optimal approximation spatial distribution for stationary nodes, and they have also developed a cooperative movement algorithm on nodes for achieving a curvature-weighted distribution pattern. This method helps obtain the crucial data for optimal environment abstraction. The authors in Ahmadi et al. (2010) have suggested congestion control mechanism for accurate estimation of spatiotemporal phenomena in wireless sensor networks performing monitoring applications. This approach emphasizes the importance of data not only by deadline but also by the dynamic state of the physical world. The design, implementation, and the evaluation of systems and algorithms that enable predictable and scalable real-time data services for real-time CPSs are investigated in Kang (2009).

However, the predictability of time and deadlines for real-time CPSs is difficult due to the complex hardware and software (e.g., programming language nature) of CPSs. Further, more requirements rather than time should be considered for the requirement engineering process of real-time CPS design.

- **Generic Requirements:** One of the critical requirements for real-time CPSs is reliability. An unreliable real-time CPS can result in system malfunctions, service disruptions, financial losses, and even human-life losses. The authors in Leon and Kaiser (2013) have proposed Failure Analysis and Reliability Estimation (FARE) framework, which is utilized for benchmarking reliability of real-time CPSs. FARE framework provides a CPS reliability model, a set of methods and metrics on the evaluation environment selection, failure analysis, and reliability estimation for benchmarking real-time CPS reliability. The requirement engineering of real-time CPSs deals with variety of stakeholders with their different objectives, which is challenging for requirement elicitation, documentation, and management. One solution to this challenge is to utilize an artifact-oriented requirement engineering approach which emphasizes on artifact and dependencies rather than processes and methods of creating the artifact. A content model based on artifact has been proposed in Penzenstadler and Eckhardt (2012), which facilitates the collaboration between stakeholders from 30 companies in ARAMiS project for real-time CPSs.

An approach on how the requirements will be collected and how the requirements are organized as well as optimized in the requirements engineering process for real-time CPSs has been developed in Rahman and Nower (2017). At the elicitation phase, this approach introduces the following requirements in the development of real-time CPSs:

1. **Business Domain Analysis:** This category of requirements comprises problem statement, mission/vision statement, objectives of the system, rationality of the approach, benefits of all relevant stakeholders, performance measurement, risk management, work plan and time line, team role, cost estimation and fund, etc. Business domain analysis renders the developer useful resources to analyze various requirements after collecting them from stakeholders.
2. **Collect System Requirements:** During the collaboration with stakeholders, system requirements are collected and listed, which include conflicting requirements, e.g., response time vs output data volume, and common requirements, e.g., system protection from other accesses and authentication. In order to optimize the various requirements and to reduce the cost and development time, this category of requirements should be managed carefully.
3. **Remove Conflict Requirements and Grouping the Requirements as:** It is necessary to remove the conflict requirements and to merge the common requirements. After optimizing the requirements, they can be refined into:
 - a. **Normal/Essential:** This type of requirements is explicitly described by stakeholders and must be met in developing the system.
 - b. **Expected:** Without this type of requirements, the system will be operating improperly and incompletely, and thus this requirement must also be met in the development of the system.
 - c. **Exciting:** This type of requirements offers extra satisfaction to clients.
4. **Cluster the Requirements:** As real-time CPSs are complex systems, containing various modules (e.g., computation, communication, and control), it is helpful to cluster the requirements according to different modules which helps distributing the workload among developers evenly. This in turn reduces the development time.

The real-time CPSs may consist of sensor networks such as wireless sensor networks (WSNs). In addition to the abovementioned requirements, in Ali et al. (2015), the authors listed the technical requirements of the CPSs, which include the following:

1. **Network Formation:**
 - a. Network formation may contain several overlapping networks with different applications.
 - b. It also supports dynamic joining and leaving of a network.
2. **Communication Pattern:**
 - a. Communication pattern supports intra-WSN communication.

- b. It also includes frequent cross-layer interaction support for control over numerous actuators and quality of service (QoS) provisioning.
- 3. **Power Management:**
 - a. CPS is supported by web cloud in most cases where power concerns may not be as critical due to abstraction of middleware network.
 - b. The monitoring station and actuation center might need to be active most of the time.
- 4. **Network Coverage:**
 - a. CPSs have broader coverage and connectivity options that lie outside of WSN domain. This is because CPSs may contain several WSNs.
- 5. **Node Mobility:**
 - a. Node mobility may include mobile and static node networks.
 - b. It may also include data from nodes of hybrid interconnected networks collected in dynamic and random fashion.
- 6. **Knowledge Mining:**
 - a. Knowledge mining encompasses more elaborate information gathering and knowledge base population.
 - b. It also includes intelligent decision-making platforms dedicated to analyzing information.
- 7. **Quality of Service:**
 - a. Compared to WSNs, QoS for CPS relates to a higher-level cross-layered approach.
 - b. In CPSs, security and confidentiality are important service aspects.

The authors in Gunes et al. (2014) have categorized the components of real-time CPSs into three main groups: physical world, interfaces, and cyber systems (Fig. 3).

The physical world is the phenomenon which is intended to be monitored or controlled. The cyber systems are embedded devices which perform computation, process information, and communicate with their distributed environments. The interfaces include communication networks and other intermediate components such as interconnected sensors, actuators, analog-to-digital converters (ADC), and digital-to-analog converters (DAC), which connect the cyber systems with the physical world. Sensors and actuators are in charge to convert other forms of energy to analogue signal (e.g., voltages) and vice versa, respectively. ADCs and DACs are in charge of converting continuous analogue signals to discrete digital signals, respectively.

In real-time CPSs, it is probable to have shared sensor and actuator networks (SANs), which makes resource scheduling a challenging task in real-time CPS operation. Therefore, it is vital to perform actuation coordination to determine which actuators must be scheduled to perform a particular action or how to manage control actions properly. When control tasks are allocated to particular actuator, several parameters including actuator capabilities, real-time guarantee, task completion time, energy consumption of each actuator, and the physical system requirements must be taken into account (Mo et al. 2014). Unlike the cyber systems which mostly have the reversibility or preemption of actuator operations (e.g., preemption

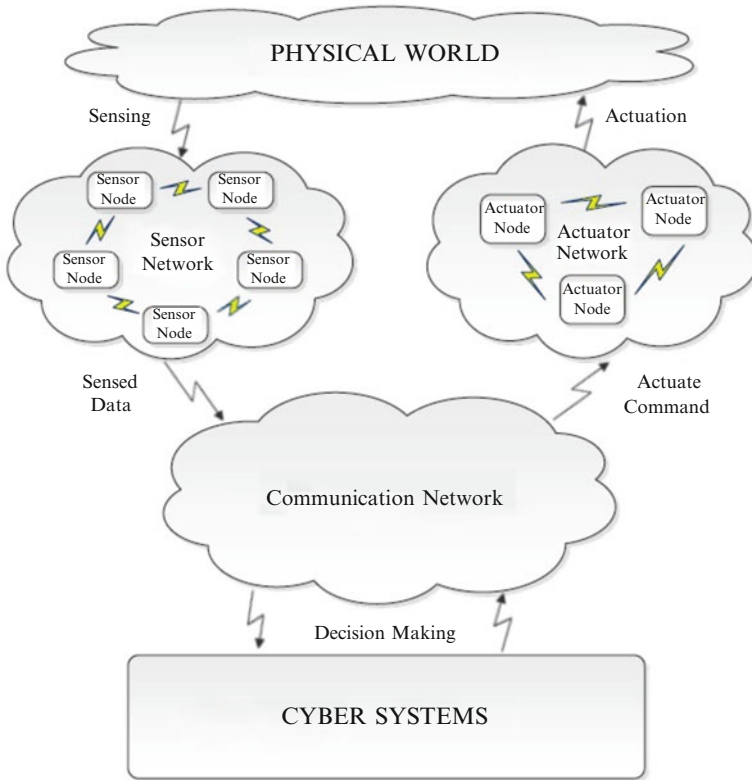


Fig. 3 Components of real-time CPS (Gunes et al. 2014)

is available for databases or bus access protocols), in real-time CPSs, the rollback operation and preemption is unavailable, i.e., physical operations executed by actuators typically are irreversible. That is, for the operations of the actuators according to erroneous data, it is usually very challenging or almost impossible to roll back the actuation performed. The specific example relevant to healthcare applications has been discussed in Luo et al. (2012). Further, irreversible operations in real-time CPSs affect real-time scheduling in the scenarios where several tasks are managed on a shared platform. Particularly, the hard real-time tasks could be blocked by low-priority processes in case a shared actuation resource access is irreversible. A specific example has been discussed in Springer et al. (2014) for a satellite communication system.

The control components are one of the essential parts in real-time CPSs. This is because the control laws of physical phenomena and the theory behind control systems are the fundamental aspects for all modern continuous time dynamical systems. For conventional systems, control policies and laws are implemented and applied after manufacturing the system prototype (Erdem et al. 2010), and thus they

are separate from the system infrastructure. However, this approach is infeasible to meet the requirements of the CPSs because they are complex and dynamic in nature. To meet the requirements of CPSs and implement complex control laws, the physical system and its dependency on control laws should be well defined and modeled (Zhou and Baras 2013).

One of the vital aspects of real-time CPSs is their real-time operations. Hence, the real-time control components are the essential parts of real-time CPSs. The conventional real-time control strategies are implemented through open-loop control, feed-forward control, and feedback control mechanism. The open-loop control mechanism needs the operator adjustment because it exploits only the input signal to actuate the output according to the control strategy and it requires the feedback mechanism to adjust the output of the system (Hopgood 2012). The feed-forward control mechanism takes into account the environmental factors which are measured by sensors in the physical system. Then, the feed-forward mechanism predicts the relationship between the physical system and its environment, and the control action is adjusted by the controller component accordingly (Hopgood 2012). The feedback control mechanism which is also known as “closed-loop” control mechanism adjusts the output signal according to the error signal between the feedback signal from the output and the input signal. As both physical system and controller component affect each other, it is called closed-loop control mechanism. All of the environmental factors which influence the physical system are considered through feedback signal (Hopgood 2012). As the physical world (i.e., physical system/environment) is one of the main components of real-time CPSs and they interact with physical systems without human intervention (via physical awareness capabilities), many of real-time CPSs are capable of utilizing both feed-forward and feedback control mechanism together.

In case the feedback control of a system is closed via a shared network, the system is called networked control system (NCS). In NCS, the control input/plant output signal is passed through interconnected components, e.g., sensors, controllers, and actuators (Gupta and Chow 2010). Supervisory control and data acquisition (SCADA) is a common control system (component). These types of control systems are used for monitoring and controlling processes such as industrial processes, infrastructure, and facility-based processes that exist in the physical world. The SCADA system collects data in real-time fashion from the remote or local sensors and sends the data to a central fusion for processing, controlling equipment/conditions, and taking the necessary actions. However, the requirements of real-time CPSs are far beyond the NCS and SCADA.

The sensor nodes and sensor networks in the sensor network technologies are closely related to some core concept in real-time CPSs. A sensor node comprises sensors, actuators, computing elements, e.g., processors, memories, communication modules, and batteries. The sensor network consists of many sensor nodes which are interconnected through wireless or wired connection (Golatowski et al. 2003). The integration of a large number of sensor nodes via wireless network connection is called wireless sensor network (WSN). In WSN, some nodes may transfer raw data to the other nodes in charge of data processing, which are called data fusion

centers. This type of WSNs consume more power and communication bandwidth. If the fusion centers fail, all the network is interrupted. In contrast, in distributed WSNs, all the nodes process the raw data by themselves, due to their computing and data processing capabilities, and relay the required part of processed data to the other sensor nodes (Zayyani et al. 2016a, b, 2017; Korki and Zayyani 2019). As the real-time operation is vital in real-time CPS, distributed WSN is beneficial for these kind of systems.

5 Related Topics to Real-Time CPSs

This section presents the various research areas and prevalent concepts, which are related to real-time CPSs. These are:

- **Big Data:** The datasets that are too large and complex to collect, store, manage (organize), and analyze with standard methods and database tools are called Big Data (Suciu et al. 2018). A large-scale real-time CPS can be considered as millions of networked nodes including smart devices, sensors, and actuators, which sense, process, and communicate data all over the network (Atat et al. 2018). The huge data transactions between these embedded networked smart devices contribute to Big Data available. Based on the number of smart devices and the size of datasets, the size of Big Data varies between multiple terabytes to many petabytes, i.e., 1024 terabytes (Sheth et al. 2013).
- **Cloud:** Cloud is a recent terminology emanating from the information and communications technology (ICT), through which the users are allowed to have an on-demand access to a shared pool of customizable computing resources such as software applications, stored information, services, hardware, etc. (Monica et al. 2013). Cloud computing enables the users to have broad network access to resources, to optimally use and control the resources, and to manage the hardware and software resources, services, etc., without human interactions. Cloud computing allows the real-time CPSs to improve their capabilities through the management and processing of aggregated sensor data using a cloud model.
- **Systems of Systems (SoS):** SoS are large and heterogeneous systems which are networked together to perform a common task. SoS comprise autonomous components with capability of independent operation and management (Lu et al. 2010; Jamshidi 2008). This terminology has been utilized by systems engineering community, which implies the concept of large-scale systems with significant economical and societal impacts, such as critical infrastructure, intelligent transportation, emergency response, etc. (Samad and Annaswamy 2011).
- **Mechatronics:** Mechatronics is a term which is comprised of two terms: “mecha” and “tronics.” Mecha refers to mechanical systems or mechanical engineering and tronics refers to electronics systems or engineering. Due to the evolution of software and information technology, mechatronics has been evolved over the last decades, although it has first been utilized in the late 1960s. Hence, mechatronics is a systematic approach to design, develop, and implement

complex engineering systems by incorporating information technologies into physical systems (Bradley and Russel 2010).

- **Cybernetics:** Cybernetics is the study of communications and control strategies in machines, systems, and living creatures. It explains the relationship between the whole system behavior and the system components (Tsien 1954). In the design of real-time CPSs, the theory of cybernetics and the practical aspects of mechatronics system design can be utilized (Suh et al. 2014).
- **Internet of Things (IoT):** The term Internet of Things (IoT) emanated from the idea of interconnecting smart devices as a network, and IoT term was coined in 1999. IoT reflects the future of radio-frequency identification (RFID) technology which allows the physical objects (or products) to be identified automatically via a small electronic chip called RFID tag. IoT helps observe and identify the things (RFID tagged objects) by capturing their data. This in turn improves the efficiency and accountability of different businesses (Ashton 2009; Caceres and Friday 2012).

IoT is closely related to real-time CPS. This is because IoT enables us to observe the things in the physical world, to exploit the communications capabilities, and to collect data which are required to control and manage the things more effectively (Ashton 2009). The original version of IoT aimed to identify and monitor various technologies; however, recently IoT targeted the control of physical systems by incorporating the RFID systems and sensor networks, which are called RFID sensor networks (Al-Fagih et al. 2012). The authors in Atzori et al. (2010) investigated different aspects of IoT, revision and improvements of its enabling technologies. Further, the idea of interconnecting heterogeneous real-time CPSs under a large-scale universal network such as the Internet has been investigated in Koubaa and Andersson (2009). This idea is called Cyber-Physical Internet (CPI).

- **Machine-to-Machine (M2M) Communication:** Machine-to-Machine (M2M) communication is a combination of smart devices including computers, embedded processors, smart sensors, actuators, and mobile devices which are connected and transferring data together through communication networks (Watson et al. 2004). M2M communication is a subcategory of IoT and real-time CPS. The architecture, standard development, and applications of M2M communication have been discussed in Chen et al. (2012). The authors in Chen et al. (2012) have also suggested to integrate intelligent road and unmanned vehicle with wireless sensor networks (WSNs) navigation as the real-time CPS. M2M communication in conjunction with IoT enables us to use new business models by rendering smart services to customers to improve efficiency and provide automation and low-cost systems in the area of e-commerce (Caceres and Friday 2012).
- **Machine Learning:** Combining the machine learning approaches and artificial intelligence methods with real-time CPS has many advantages. Machine learning methods can be utilized in control and monitoring system of real-time CPSs for detection, control, execution, and communication (Lv et al. 2021). For instance, the deep neural network (DNN) can be used to enhance the level of autonomy in real-time CPSs (Cai et al. 2020). Also, for improving the security of real-

time CPSs, deep learning-based methods can be used for anomaly detection (Luo et al. 2021). Further, the generative adversarial networks (GANs) can be used for fault and cyberattack detection in real-time CPSs by dimensionality reduction (Farajzadeh-Zanjani et al. 2021).

Among the abovementioned concepts, IoT and real-time CPS are the two major disruptive civil technologies with potential impacts on the US interest (Disruptive Civil Technologies 2008). Real-time CPS and IoT are the two emerging technologies which can represent an automatic network of connected sensors, actuators, and hardware devices (Ur Rehman and Gruhn 2018). The next-generation Internet technologies will have great impact on IoT and CPS research areas. In the way the conventional Internet revolutionized our interactions with each other, the IoT and real-time CPS will revolutionize our interactions with physical world (Paul et al. 2011).

6 Applications of Real-Time CPSs

Due to their unique characteristics, real-time CPSs have been utilized in many domains (Khaitan and McCalley 2015). In the sequel, we briefly review some of these applications.

6.1 Intelligent Manufacturing

Intelligent manufacturing (IM) is deploying the embedded software and hardware technologies to optimize the manufacturing of goods and service delivery (Koziolek et al. 2011; Yao et al. 2019). Intelligent factory is another terminology which is used for next-generation IM. One of the major applications of real-time CPSs is the IM. This is because mass production, domestic and international marketing, and economic growth are all parts of IM. In Europe and the USA, CPSs have been characterized for IM. The Industry 4.0 project started in Germany, which represents a major opportunity for manufacturing of the future. The Industry 4.0 is provisioned to be the leading in manufacturing of the future. The core of Industry 4.0 is real-time CPS to achieve intelligent factory. The dynamic configuration of production is the major component of Industry 4.0. Unlike the traditional production methods, the dynamic configuration in the manufacturing process is capable of changing the original design at any time. In the intelligent factory of Industry 4.0, the fixed production line has been replaced by the dynamically modular production. The Industry 4.0 is aimed to improve the automation, operational efficiency, and effectiveness. The virtual reality, artificial intelligence, industrial Internet, industrial Big Data, industrial robot, 3D printing, cloud computing, knowledge work automation, and industrial network security are essential technologies which have been utilized in Industry 4.0. Further, these technologies will create various business opportunities.

The Industry 4.0 includes a set of future industrial developments such as cyber-physical systems (CPSs), the Internet of Things (IoT), the Internet of Services (IoS), robotics, Big Data, cloud manufacturing, and augmented reality. The utilization of these technologies helps develop more IM processes, which include devices, machines, production modules, and products that are able to independently exchange information, trigger actions, and control each other, thus enabling an IM environment (Nagi et al. 2018).

In future, the manufacturing and production will experience a high demand of flexibility. However, the safety makes it very challenging to meet the requirements of these demands. The lack of safety is due to the close interaction between machines and human experts without adequate sensors and intelligent devices to avoid possible accidents (Gide 2013). Real-time CPSs in intelligent manufacturing will enhance safety, productivity, and efficiency by integrating embedded system production technologies to increase the flexibility of work flow and new form of collaboration (Trade&Invest 2013).

6.2 Vehicular Systems and Intelligent Transportation

Almost all modern vehicles are real-time CPSs, which include advanced navigation systems, displays, and entertainment and manage the motion and the energy consumption of the vehicle. The functionality of real-time CPS has been utilized in various aspects of vehicular system including data fusion in distributed CPS, public transport, design of cyber-physical vehicles, electrical vehicle charging, and road monitoring.

The AUTomotive Open System ARchitecture (AUTOSAR), which offers real-time cyber infrastructure in modern automotive platform, has been introduced in Bhatia et al. (2010). AUTOSAR meets the system analysis and implementation, which are the requirements of the system development. AUTOSAR is a modular system including the software components, operating system, and communication model. The human-centric data fusion in distributed vehicular CPS has been discussed in Wagh et al. (2011). This model is a design architecture that considers human factors and employs safety applications to improve the safety of the drivers. This model includes a fusion center to fuse and process multiple messages and extract useful information from those messages. The advantages of this design include preventing positional negative effects of cyber applications on driver, e.g., information overload, confusion, and distraction. A system performance optimization model for unmanned vehicle CPS with wireless sensor network (WSN) navigation has been introduced in Yan et al. (2012). The vehicle receives the collected data from WSN to determine the direction of movements. To increase the accuracy of positioning, the real-time, fast communication and response of the vehicle is essential. Toward this end, the particle swarm optimization algorithm is used in this model. The design of a system for facilitating public transportation has been discussed in Lau et al. (2011). This system, which is called ContriSenseCloud, is based on client-server model, in which the data is collected and processed by

public masses. This system comprises interactive participatory sensor network, which enables users to collect, analyze, and share local knowledge. Hence, the journey plan of the users is formed by the information distributed from user-contributed data.

Intelligent transportation system (ITS) deploys advanced technologies such as sensing, communication, computation, and control mechanisms in transportation systems to enhance safety, coordination, and services in traffic management with real-time information sharing. ITS helps facilitate both ground and sea transportation through information sharing over satellites. It further provides a communication network between vehicles, the infrastructure, and passengers' portable devices. ITS connects pedestrians, vehicles, sensors, roadside infrastructures, traffic management centers, satellites, and other transportation system components by utilizing variety of wireless communication technologies and standards (Qu et al. 2010). In addition, the next-generation ITSs enable the authorities to monitor real-time traffic; to increase safety in transportation and comfort through information exchange among traffic users; to optimize traffic management; to prevent collision; and to utilize the satellite-based technology to connect drivers, roads, and vehicles smoothly. With the integration of real-time CPS into infrastructures, vehicles, and roadways, ITSs can achieve driver assistance, collision avoidance or notification, improvements in travel time without fear of unexpected delays, reductions in congestion, and advanced control over infrastructure and vehicles for energy saving (Networking and Information 2011). ITSs benefit from both advanced sensor and embedded computer systems technology, and wireless, cellular, and satellite technologies for vehicle-to-vehicle (V2V), vehicle-to-pedestrian (V2P), and vehicle-to-infrastructure (V2I) communication to better manage complex traffic flow, ensure safety, and extend situational awareness.

6.3 Medical and Healthcare Systems

Medical and healthcare systems address different aspects of the patient's physiology, in which real-time CPSs provide important roles. Medical applications of real-time CPSs include implantable/life support medical devices, robot-assisted operation, development of medical application platform, technologies related to home care, assisted living, smart operating room, smart medical devices (e.g., pacemaker, medical ventilator, infusion pump, etc.), and smart prescription (Baheti and Gill 2011; Kim and Kumar 2012). The challenges and research trends in the design and development of the real-time Cyber-Physical Medical System (CPMS) are discussed in Sokolsky (2011). These research trends demonstrate an increase in developing reliable software-based systems, connectivity of medical devices equipped with network interfaces, and continuous demand for patient monitoring such as home care, assisted living, telemedicine, sport activity monitoring, etc. It is worth noting that modeling and model-driven engineering has a great impact on the development of real-time CPMS (Sokolsky 2011). An example for real-time CPMS is the Medical Device Plug-and-Play (MD PnP) (Goldman et al. 2005), which provides

a framework for a safe interconnectivity and interoperability of medical devices. The available interoperability in MD PnP helps increase patient safety, enable new treatment options and proliferation of technology, and improve existing medical practice. The real-time CPSs can be utilized in the design of healthcare systems for disabled people and frail elderly people (Lim et al. 2011), in the development of controlled artificial legs (Huang et al. 2010), and in the design and analysis of body area networks (BAN) (Banerjee et al. 2012a). The next-generation CPMSs should render extensive data integration and access, comprehensive data acquisition and analysis, closed-loop control capabilities, energy efficiency, real-time visualization, and plug-and-play capability with interoperable medical devices.

6.4 Smart Grid

Smart grids refer to electric networks that deploy advanced monitoring, control, and communication technologies to deliver reliable and secure energy supply, improve operational efficiency of generators and distributors, and provide flexible choices for consumers. Smart grids are comprised of complex network systems and cyber-physical systems, which encounter several technological challenges. The smart grids introduce the transformation from a centralized, producer-controlled electric grid to distributed, cooperative, responsive, and consumer-interactive grid by integrating future information and communication technologies for grid modernization (Farhangi 2010). The advantages of smart grids include real-time load monitoring, distribution, and planning at utility level; a balance of supply and demand at the device level; two-way flow of information (i.e., real-time communication between the consumer and utility); the integration of existing energy resources into the grid; large-scale grid awareness and ability to switch between high-level (e.g., state-wide) and low-level (e.g., street-wide) grid exploration; real-time integration of sensor data with geographical information; power quality and blackouts monitoring; and prevention or minimization of a potential outage. The smart grids demonstrate the typical features of real-time CPSs due to integration of the physical systems (power network infrastructure) and cyber systems (sensors and ICT). The typical real-time CPS-based features of smart grids include (Zhao et al. 2010):

- Real-time integration of real and virtual worlds in a flexible and dynamic environment, where the physical systems scenarios as the input are supplied to the control center of CPSs, to enhance the performance of simulation models in the prediction of physical system behavior in future times
- Flexible and dynamic connections and interactions between physical and cyber systems via ad hoc communication networks with timely responses as their dynamic cooperation
- Requirement of real-time parallel computation and distributed information processing of Big Data and data streams to improve the time response of smart grid operation to transient, distribution, and scheduling layers through the CPS

- Self-configuration, self-adaption, self-organization, and self-learning which reduce the CPS response time to faults, attacks, and emergencies to improve the smart grid resilience, and security and safety of energy supply

The components of cyber technologies including communication networks and sensing devices are unchangeable on power systems and incapable of adapting to suit power system features. Hence, in order to meet the safety and security requirements of smart grids, they need calibrations and patching fixtures.

The applications of real-time CPSs in electric power grid and energy systems include enhancing the sustainability of power system by optimizing resource utilization and management; modeling the complex interactions between physical and cyber components, i.e., electric grids; and monitoring infrastructure, maintaining the security of the energy system against cyber threats. The security includes detection, prevention, mitigation, and restoration. The current cyber security research in energy system includes mostly vulnerability/attack detection and prevention. However, mitigation of the attacks and restoration of the energy systems are also important. In Saber and Venayagamoorthy (2010), the authors proposed an integrated electricity and transportation infrastructure for utilizing renewable energy sources (RESs). They presented the energy system as a CPS, where the physical components include RESs, gridable vehicles (GVs), and thermal power plants. The cyber component is an onboard system in a GV, which communicates with utility and vehicle owner's preferences, for better functionality. The resource utilization using CPS-based method has been described in Zhang et al. (2009). The authors proposed a dynamic model of battery to demonstrate the capacity behavior of a battery under time-varying discharge current. In this CPS-based model, the trend of the drawn current from the batteries is determined by the control strategies and online scheduling algorithms, based on the processor and state of the plant. Therefore, the battery model proposed in Zhang et al. (2009) determines the optimal discharge profile for a square wave current. An effective model that represents a CPS-based energy system has been suggested in Ilic et al. (2010). All the physical components of this model are modules interconnected by an electric network. In this CPS-based model, all components are represented as cyber-physical module, which are all specified by the cyber and physical input–output signals, internal dynamics, local sensing, and actuation. It is worth noting that the modular components are integrated to the CPS-based system according to network topology and constraints.

6.5 Smart Buildings

Smart building is based on utilizing various sensors, actuators, and distributed control systems to provide optimum control and automation of heating, ventilation, and air-conditioning (HVAC), lighting, fire prevention, and security systems in the buildings. Smart building is the complement for smart grid and smart city concepts. IoT and real-time CPSs introduce new applications in the smart buildings through various smart building appliances such as entertainment media which

require interaction patterns for realizing such systems (Chatzigiannakis et al. 2014). The next-generation smart buildings are expected to connect the smart grids with smart living environments and predicting their living patterns to ensure comfortable living environments. Smart networks including home area networks (HANs), neighborhood area networks (NANs), and wide area networks (WANs) can be utilized in the next-generation smart power distribution network to ensure two-way flow of electricity and information. HAN in smart buildings can deliver data traffic and control instructions between both the smart utilities (e.g., smart meters) and the residents' smart devices, and the residents' smart devices themselves (Meng et al. 2014).

6.6 Data Centers

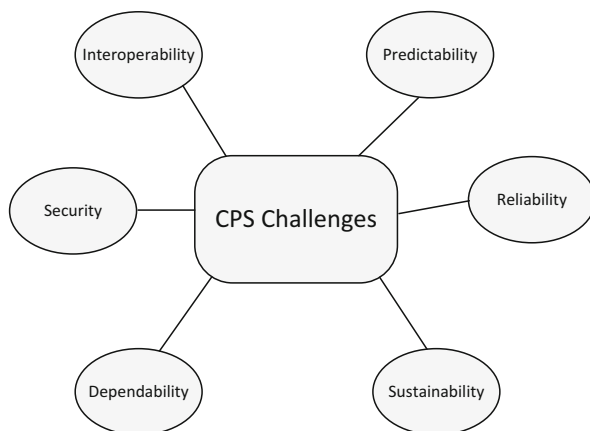
A data center can be modeled as a real-time CPS. This is because the data center includes online applications and services to ensure correct functioning. The complicated interaction of cyber and physical components in huge data centers requires the principles of real-time CPS design to resolve the issues in data center management. The authors in Parilini et al. (2010) proposed an approach for data centers to optimize the quality of computational services while keeping the energy costs for computation and cooling minimal. The proposed model includes the interaction between cyber components, i.e., the computational network representing the distribution and flow of computational tasks, and physical network component, i.e., the thermal network characterizing the distribution of thermal energy. Cyber and physical systems have close interactions in data centers. For instance, work-scheduling algorithms affect temperature distribution. In Li et al. (2011), the authors proposed a temperature predicting model to predict the temperatures near the servers in a data center based on continuous streams of temperature and airflow measurements. The approach, which utilizes physical laws and sensor observations in the data center, uses the data obtained from the sensor to learn the parameters of a data center's CPS.

7 Challenges of Real-Time CPSs

Real-time CPSs transform the human interaction with physical world. However, this transformation incurs some costs. There exist several significant challenges for implementing real-time CPSs, which should be addressed to facilitate adoption of real-time CPSs in real-world scenarios. This section presents the most important system-level challenges. Figure 4 shows the main real-time CPS challenges. In the sequel, we present an overview of the challenges in the real-time CPSs.

1. **Dependability:** The specific property of a system to function without significant degradation in its performance is called dependability. The degree of the reliability of the whole system is described by its dependability. A system

Fig. 4 Real-time CPS challenges (Gunes et al. 2014)



which operates without intrusion and failure and delivers the required services accurately is a highly dependable system. Before the actual system operation, it is very difficult to ensure whether the system is dependable. For instance, time delay in sensor data reading and actuator operation may degrade dependability and result in unpredictable consequences. The physical and cyber components of the real-time CPSs are inherently interdependent. As the underlying components are likely interconnected dynamically during system operation, the analysis of the dependability becomes very difficult. To resolve this issue, common terminologies and semantics for describing dependability-related information across constituent systems/underlying components should be developed in the design stage of the real-time CPSs (Denker et al. 2012; Guan et al. 2020; Xu et al. 2020). The attributes of dependability include:

- **Availability:** This property of a CPS should ensure the readiness of the system even during the faulty conditions. According to this property, a highly available system is capable of isolating malfunctioning portion from itself and properly operating without it. Destructive cyberattacks such as denial of service attacks significantly hinder the system availability. For instance, in real-time Cyber-Physical Medical Systems, medical data is of highly importance, on which necessary action for saving patient's life should be taken. Hence, destructive attacks or system/component failure leads to unavailability of such data, which in turn poses risk on the patient's life (Haque et al. 2014).
- **Safety:** Safety is an important feature of a system, which prevents any harm, hazard, or risk inside or outside of the system during its operation. A highly safe system must comply the safety regulations and employ safety assurance mechanisms in fault condition. In intelligent manufacturing, real-time tracking of sustainable production and real-time management of processes results in improved safety. Automation of process control via embedded control systems and real-time data collection via sensors (across

the manufacturing enterprise) will highly improve the safety of manufacturing plants. Implementation of wireless sensor networks will help in the detection of operational failures and prevention of catastrophic consequences.

2. **Reliability:** This property refers to the degree of the correctness in the functionality of the system. As real-time CPSs must operate reliably in open and uncertain environments with uncertain knowledge, attribute (e.g., timing), or outcome of a process, it is essential to quantify uncertainties during CPS design stage. Thus, the uncertainty analysis renders real-time CPS efficient reliability features. In addition, the accurate functionality of physical and cyber components, potential errors in the design, and ad hoc cross-domain network connections reduce the real-time CPS reliability. The reliability of a real-time CPS is described as follows:

- **Robustness:** Robustness is an important feature of a system which describes the capability of the system to maintain its stable configuration and withstand any failure. Therefore, a highly robust system should maintain its functionality in the presence of any failure without major changes to its original configuration. Further, a highly robust system avoids any hindrance in its operation due to any failures. In addition, possible disturbances emanating from sensor noises, actuator inaccuracies, faulty communication channels, potential hardware errors, or software bugs may reduce the overall robustness of real-time CPS. Several factors such as modeling integrated system dynamics (e.g., actual ambient conditions in which CPSs operate), evolved operational environment, or unforeseen events may affect the run time of real-time CPSs and thus should be considered in the design of the robust real-time CPSs (Rungger and Rungger 2016; Finkbeiner et al. 2021).

3. **Predictability:** Predictability is the qualitative or quantitative degree of predicting state/behavior/functionality of a system. A highly predictable system should guarantee the specific behavior/functionality of the system in real-time operation, while all system requirements are met. For instance, in real-time Cyber-Physical Medical Systems, intelligent medical devices with advanced control technologies are expected to adjust themselves based on patient's conditions, predict patient's movements, and change their features based on the conditions of the surrounding environment. Although many intelligent medical devices operate in real time and demonstrate various sensitivity to timing uncertainties (e.g., delays, jitters, etc.), some components of real-time CPMS are unpredictable in time. Hence, along with effective programming and networking abstractions, new resource allocation and scheduling strategies should be developed to guarantee predictable end-to-end timing constraints (Mubeen et al. 2020). The following attributes describe the predictability of a real-time CPS:

- **Accuracy:** Accuracy is a metric to compare the degree of closeness of a measured/observed outcome of a system to its actual/calculated outcome. A highly accurate system is capable of obtaining the actual outcome which is very close to the observed or measured outcome. In particular, for real-time CPS applications, where even slight inaccuracies may lead to system failure, the high accuracy becomes of great importance. For instance, in a real-time

motion-based object tracking system, if the sensor operates imperfectly, it will result in incorrect object position estimation, leading to the system failure (Gunes et al. 2013).

- **Compositionality:** How well the nature of a system can be understood by investigating every constituent of it is determined by compositionality of the system. In a highly compositional system, the behavior of its components provides the detailed behavior of the whole system. Designing a highly compositional real-time CPS is very challenging because the behavior of underlying physical components is chaotic. Thus, in designing a compositional real-time CPS, it is required to analyze the behavior of all cyber and physical components and devise cyber-physical methodologies for assembling CPSs from individual cyber and physical components. Further, it is essential to design standard test benches for the evaluation of all the components, and holistic mathematical model of the overall system and its components.
- 4. **Sustainability:** The capability of a system in surviving without compromising its requirements and concurrently renewing its resources and utilizing them efficiently refers to sustainability. A highly sustainable system is an enduring system with self-healing and dynamic tuning properties under evolving conditions. Sustainability is an important feature in real-time CPSs such as energy systems in terms of energy provision and management policies. For instance, the smart grid provides energy distribution, management, and customization by integrating the renewable energy sources (RESs). However, the intermittent properties of RESs and unknown load characterization impede the long-term sustainability of the smart grid. To maintain sustainability, it requires planning and operation, real-time CPS performance measurements, dynamic optimization techniques for energy management, and designing and developing autonomous micro grids (Banerjee et al. 2012b; Momoh 2010). Adaptability, resilience, reconfigurability, and efficiency are the major attributes of the sustainability of a real-time CPS:
 - **Adaptability:** The capability of a system to adjust its state to endure by changing its own configuration against different circumstances in the environment is called adaptability of the system (Audrito et al. 2021). A highly adaptable system is capable of quickly adapting to evolving requirements/situations. For instance, adaptability is one of the important features in the next-generation air transportation systems. In the next-generation air transportation, the airspace performance is improved by its automatic air transportation network, which enables air vehicles to accommodate themselves to evolving operational environment such as weather conditions, air vehicle routing and other pertinent flight trajectory patterns over satellites, air traffic congestion, and issues related to security (Agarwal 2012).
 - **Resilience:** If a system is capable of keeping its quality operation and delivery of service in the exposure to any disturbance (which is well below its endurance limit), e.g., sudden defect, malfunctioning components, rising workload, etc., it is called a resilient system. A highly resilient system has the properties of self-healing, early detection, and fast recovery in case failures occur. The resilience capability is of great importance in safety-

critical systems such as automated brake control in vehicular CPS, air and oxygen flow control over an automated medical ventilator, etc. Safety-critical CPSs are often required to operate even if some components are disrupted. Therefore, designing a perfect resilient real-time CPS needs comprehensive contingency analysis, the resilience features of the relevant application, and system evolution as a result of dynamic nature of operational environment (Denker et al. 2012; Kim et al. 2021).

- **Reconfigurability:** If a system is capable of changing its own configuration in the context of a failure or on request, it is called reconfigurable system (Isern et al. 2020). However, to be a highly reconfigurable system, it should be self-configurable, i.e., capable of dynamic fine-tuning itself. A real-time CPS can be considered as an autonomously reconfigurable engineered system. In some real-time CPS applications, real-time and remote monitoring, and control mechanism are essential, e.g., international border monitoring, wildfire emergency management, gas pipeline monitoring, etc. According to the specific application, the operational requirements such as security threat level updates, regular code updates, efficient energy management, etc. may change in such applications, which require substantial reconfiguration of the deployed sensor/actuator nodes or even the entire network to comply with the quality of service (QoS) regulations (Sanjay and Eronu 2012).
 - **Efficiency:** The amount of resources such as energy, cost, time, etc., which is required by a system to provide specific functionalities, refers to efficiency (Bayhan et al. 2020). A highly efficient system is capable of operating appropriately with optimum amount of resources. Efficiency is of great importance for energy management in real-time CPS applications like smart buildings. For instance, smart buildings can turn off HVAC (heating, ventilation, and air-conditioning) units if the building is detected as vacant to save more energy. In addition, the smart buildings can provide automatic pre-heating or pre-cooling services according to the occupancy prediction techniques (Scott et al. 2011).
5. **Security:** Security is one the important features of a system, which controls access to the system components and resources and monitors sensitive information and protects them from unauthorized access. Thus, a highly secure system offers sophisticated protection mechanism against unauthorized modification of information and unauthorized withholding of resources and should prevent the disclosure of sensitive information (Humayed et al. 2017; Olowononi et al. 2021). Some properties of real-time CPSs, such as scalability, complexity, and dynamic characteristic, make them vulnerable to cyber and physical attacks and failure (Ding et al. 2018; Li et al. 2015; Mitchell and Chen 2016). Eavesdropping, denial of service, and injecting fake sensor measurements or actuation requests related to cyber or physical components to disrupt the system operation or to steal sensitive information are some examples of malicious attacks. Utilizing large-scale networks (e.g., power grid or Internet), using unreliable communication protocols, and rapid adoption of commercial off-the-shelf (COTS) technologies are also the major factors which cause security threats to real-time CPSs (Anwar and Ali 2012). The security is attributed by:

- **Integrity:** Integrity is a feature of a system to protect itself and the information within it from unauthorized modification to maintain the accuracy of the information. A system with high integrity offers hierarchy level of authorization and consistency check mechanisms. Integrity is an essential feature of a real-time CPS. It is important to develop effective integrity check mechanisms for real-time CPSs in network packets, distinguishing malicious behaviors from the ambient noise, identifying false data injection and compromised sensor/actuator components, etc. Toward this end, it is important to investigate the characteristics of both cyber and physical components of real-time CPSs to develop a high integrity assurance (Mo and Sinopoli 2012).
 - **Confidentiality:** Confidentiality of a system only allows authorized parties to access classified and sensitive information within the system. Therefore, a highly confidential system employs very secure mechanisms and protocols to protect itself from unauthorized access, disclosure, or tampering. In real-time CPS applications, data confidentiality is of great importance. For instance, in emergency sensor network, attacks on confidential transmitted data may disrupt the emergency system or may degrade the performance of the emergency system. Sensitive information and confidential data which are transmitted via attacked sensor nodes can cause potential eavesdropping of critical data and can create fake node identities to be generated in the network. In addition, false/corrupted data can flow in the network over the fake nodes. Thus, the confidentiality of data over the network should be kept in an appropriate level (Loukas et al. 2013; Hou et al. 2020).
6. **Interoperability:** The capability of the systems to interact with each other, exchange data/information, and use this data/information to deliver services refers to interoperability. A highly interoperable system is capable of supplying or accepting services via communications and interoperation among components (Fatima et al. 2020). Unmanned aerial vehicles (UAVs) are among those systems that calls for seamless communication between each other and various ground operating vehicles (Korki et al. 2019). However, the lack of interoperability standards leads to ineffective complicated and critical missions. In smart grids, also dynamic standard should be developed to guarantee the interoperability of its components. Interoperability has the following attributes:
- **Composability:** The capability of merging different components within a system and interrelationships refers to composability (Tolk 2020). If a given system is capable of recombining its components repeatedly to meet the specific requirements, it is called highly composable. Composability should be investigated in different aspects such as device composability, code composability, service composability, and system composability. Among these aspects, system composability is more challenging, and thus it needs to appropriately define composition methodologies that follow composition features. Further, requirements, tests, and evaluations should follow composition properties. Therefore, it is of great importance to incrementally add emerging systems/components to next-generation real-time CPSs without degrading the operation of the resulting system.

- **Heterogeneity:** Heterogeneity is an important feature of a system to form a complex whole by integrating a set of various interacting and interconnected components (Keshk et al. 2021). Real-time CPSs are intrinsically heterogeneous due to constituent physical dynamics, computational elements, control logic, and deployment of diverse communication technologies. Thus, real-time CPSs require heterogeneous composition of all system components. Future medical devices as real-time CPSs are likely to be interconnected in complex open systems with a plug-and-play property and thus necessitate a heterogeneous control network and closed-loop control of interconnected devices. However, those kinds of medical devices have a highly dynamic configuration according to patient-specific medical considerations. The ever increasing emerging technologies transform the next-generation medical systems to the systems that offer situation-aware component autonomy, cooperative coordination, real-time guarantee, and heterogeneous personalized configurations more complex than that of current medical systems.
- **Scalability:** The capability of a system to continue its operation even when its size or workload is increased/decreased is called scalability (Zou 2021). Accordingly, the increase in system resources should lead to increase in the system throughput. A highly scalable system is capable of distributing and collecting mechanisms to balance its workload and finding effective communication protocols to enhance its performance. In real-time CPSs, scalable embedded components with programmable interconnected network can be employed to compensate for increasing computation demand in real-time CPSs (Gunes and Givargis 2014). In addition, in order to enable the real-time CPS components to join and leave the existing network dynamically, a high performance and highly scalable infrastructure is essential. Toward this end, dynamic software updates facilitate the dynamic update of real-time CPS applications and make utilizing real-time CPS resources more productively (Park et al. 2010).

In general, the challenges/attributes of real-time CPSs are related to each other. In Fig. 4 we highlight the overall view of system requirements/challenges for real-time CPS applications. However, real-time CPSs are a broad area of research and thus have various applications with different scales. All of real-time CPS applications are required to be designed and developed taking into account their impact on real-world and essential system-level challenges/requirements.

8 Conclusions

The real-time cyber-physical system (CPS) is a promising model for the design and development of current and next-generation engineered systems and is anticipated to have significant impact on human interaction with physical world. The real-time

CPS replaces the idea of independent cyber or physical system with integrated system design using computation and communication components. This chapter presents the relevant terminology, related topics, significant challenges, practical applications, and dominant research domains of real-time CPSs. Due to its broad area of research, real-time CPSs span diverse applications in different scales. For each application unique challenges and the integration of modern technologies into relevant application are essential. We have also highlighted that the existing systems include limited perception of the real-time CPS requirements, and extensive design and development approaches are essential to achieve the overall system objectives.

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